2 Algebra

What students need to learn:

Ref	Content descriptor	Concepts and skills
A a	Distinguish the different roles played by letter symbols in algebra, using the correct notation	• Use notation and symbols correctly
A b	Distinguish in meaning between the words ‘equation’, ‘formula’ and ‘expression’	• Write an expression • Select an expression/equation/formula from a list
A c	Manipulate algebraic expressions by collecting like terms, by multiplying a single term over a bracket, and by taking out common factors	• Manipulate algebraic expressions by collecting like terms • Multiply a single algebraic term over a bracket • Write expressions using squares and cubes • Use simple instances of index laws • Factorise algebraic expressions by taking out common factors • Write expressions to solve problems • Use algebraic manipulation to solve problems
A d	Set up and solve simple equations	• Set up simple equations • Rearrange simple equations • Solve simple equations • Solve linear equations, with integer coefficients, in which the unknown appears on either side or on both sides of the equation • Solve linear equations which include brackets, those that have negative signs occurring anywhere in the equation, and those with a negative solution • Solve linear equations in one unknown, with integer or fractional coefficients

<table>
<tr><th>Ref</th><th>Content descriptor</th><th>Concepts and skills</th></tr>
<tr><td>A f</td><td>Derive a formula, substitute numbers into a formula and change the subject of a formula</td><td>Derive a simple formula, including those with squares, cubes and roots
Use formulae from mathematics and other subjects expressed initially in words and then using letters and symbols
Substitute numbers into a formula
Substitute positive and negative numbers into expressions such as $3x^2 + 4$ and $2x^3$
Change the subject of a formula
(NB: Rearranging of formulae using square roots or squares is Higher Tier only)</td></tr>
<tr><td>A g</td><td>Solve linear inequalities in one variables, and represent the solution set on a number line</td><td>Solve simple linear inequalities in one variable, and represent the solution set on a number line
Use the correct notation to show inclusive and exclusive inequalities</td></tr>
<tr><td>A h</td><td>Use systematic trial and improvement to find approximate solutions of equations where there is no simple analytical method of solving them</td><td>Use systematic trial and improvement to find approximate solutions of equations where there is no simple analytical method of solving them</td></tr>
<tr><td>A i</td><td>Generate terms of a sequence using term-to-term and position-to-term definitions of the sequence</td><td>Recognise sequences of odd and even numbers
Generate arithmetic sequences of numbers, squared integers and sequences derived from diagrams
Write the term-to-term definition of a sequence in words
Find a specific term in a sequence using position-to-term or term-to-term rules
Identify which terms cannot be in a sequence</td></tr>
<tr><td>A j</td><td>Use linear expressions to describe the n^{th} term of an arithmetic sequence</td><td>Find the n^{th} term of an arithmetic sequence
Use the n^{th} term of an arithmetic sequence</td></tr>
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Foundation

Ref	Content descriptor	Concepts and skills
A k	Use the conventions for coordinates in the plane and plot points in all four quadrants, including using geometric information	• Use axes and coordinates to specify points in all four quadrants in 2-D • Identify points with given coordinates • Identify coordinates of given points (NB: Points may be in the first quadrant or all four quadrants) • Find the coordinates of points identified by geometrical information in 2-D • Find the coordinates of the midpoint of a line segment • Calculate the length of a line segment
A l	Recognise and plot equations that correspond to straight-line graphs in the coordinate plane, including finding gradients	• Draw, label and scale axes • Recognise that equations of the form $y = mx + c$ correspond to straight-line graphs in the coordinate plane • Plot and draw graphs of functions • Plot and draw graphs of straight lines of the form $y = mx + c$ • Find the gradient of a straight line from a graph
A r	Construct linear functions from real-life problems and plot their corresponding graphs	• Draw straight line graphs for real-life situations – ready reckoner graphs – conversion graphs – fuel bills – fixed charge (standing charge) and cost per unit • Draw distance-time graphs (NB: Quadratic functions from real life problems are at Higher Tier only)

Ref	Content descriptor	Concepts and skills
A s	Discuss, plot and interpret graphs (which may be non-linear) modelling real situations	• Plot a linear graph • Interpret straight-line graphs for real-life situations – ready reckoner graphs – conversion graphs – fuel bills – fixed charge (standing charge) and cost per unit • Interpret distance-time graphs • Interpret information presented in a range of linear and non-linear graphs
A t	Generate points and plot graphs of simple quadratic functions, and use these to find approximate solutions	• Generate points and plot graphs of simple quadratic functions, then more general quadratic functions • Find approximate solutions of a quadratic equation from the graph of the corresponding quadratic function